

Stochastic Gradient Descent for Strongly Convex and Smooth Functions

1 Algorithm Description

The algorithm iteratively updates the parameters by computing gradients on randomly sampled minibatches and taking a step in the negative gradient direction with a fixed step size. It is designed for strongly convex and smooth objective functions.

1.1 Mathematical Formulation

The algorithm is a variant of Stochastic Gradient Descent (SGD) designed for optimizing strongly convex and smooth objective functions. The goal is to minimize an objective function $f : \mathbb{R}^d \rightarrow \mathbb{R}$, which is both strongly convex and smooth. A function f is strongly convex with parameter $\mu > 0$ if for all $\mathbf{w}, \mathbf{v} \in \mathbb{R}^d$,

$$f(\mathbf{v}) \geq f(\mathbf{w}) + \nabla f(\mathbf{w})^T(\mathbf{v} - \mathbf{w}) + \frac{\mu}{2}\|\mathbf{v} - \mathbf{w}\|^2$$

and is L -smooth if for all $\mathbf{w}, \mathbf{v} \in \mathbb{R}^d$,

$$\|\nabla f(\mathbf{v}) - \nabla f(\mathbf{w})\| \leq L\|\mathbf{v} - \mathbf{w}\|.$$

Update Rule: The algorithm updates the parameters $\mathbf{w}$ using the gradient computed on a minibatch:

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla f(\mathbf{w}_t)$$

where:

- $\mathbf{w}_t$ is the parameter vector at iteration t .
- $\eta > 0$ is a fixed step size (learning rate).
- $\nabla f(\mathbf{w}_t)$ is the gradient of the objective function with respect to $\mathbf{w}_t$.

Hyperparameters:

- η : Step size controls the rate of convergence. For convergence, it should satisfy $0 < \eta < \frac{2}{L}$, where L is the Lipschitz constant of the gradient.

- Minibatch size: Number of samples used to compute the gradient approximation in each iteration. For deterministic functions, this is typically 1.

2 Pseudo Code and Dry Run

2.1 Pseudocode

Algorithm StochasticGradientDescent

Input: Objective function f , initial parameter w_0 , step size η , max iterations T

Output: Optimal parameter w^*

```
Initialize  $w = w_0$ 
for  $t = 0$  to  $T-1$  do
    Compute gradient:  $g = \nabla f(w)$ 
    Update parameters:  $w = w - \eta * g$ 
    if convergence criteria is met then
        break
end for
return  $w$ 
```

2.2 Dry Run Example

Consider the quadratic function $f(w) = (w - 3)^2$. The gradient is $\nabla f(w) = 2(w - 3)$.

Initialization:

- $w_0 = 0$
- Step size $\eta = 0.1$
- Minibatch size = 1

Iteration 1:

- Compute gradient at w_0 : $\nabla f(w_0) = 2(0 - 3) = -6$
- Update $w_1 = w_0 - \eta \nabla f(w_0) = 0 - 0.1 \times (-6) = 0.6$
- Objective value: $f(w_1) = (0.6 - 3)^2 = 5.76$

Iteration 2:

- Compute gradient at w_1 : $\nabla f(w_1) = 2(0.6 - 3) = -4.8$
- Update $w_2 = w_1 - \eta \nabla f(w_1) = 0.6 - 0.1 \times (-4.8) = 1.08$
- Objective value: $f(w_2) = (1.08 - 3)^2 = 3.6864$

Iteration 3:

- Compute gradient at w_2 : $\nabla f(w_2) = 2(1.08 - 3) = -3.84$
- Update $w_3 = w_2 - \eta \nabla f(w_2) = 1.08 - 0.1 \times (-3.84) = 1.464$
- Objective value: $f(w_3) = (1.464 - 3)^2 = 2.359296$

This example demonstrates how the parameters converge towards the optimal value with each iteration.

3 Convergence Theory

3.1 Assumptions

We consider the optimization problem of minimizing a strongly convex and smooth function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ using a stochastic gradient descent algorithm. Our assumptions are as follows:

1. **Strong Convexity:** The function f is μ -strongly convex, meaning there exists $\mu > 0$ such that for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2.$$

2. **Smoothness:** The function f is L -smooth, meaning there exists $L > 0$ such that for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$:

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2.$$

3. **Gradient Noise:** The gradients computed using minibatches are unbiased estimates of the true gradient with bounded variance σ^2 . Formally, if $\nabla f(\mathbf{x}, \xi)$ is the gradient estimate using minibatch ξ , then $\mathbb{E}[\nabla f(\mathbf{x}, \xi)] = \nabla f(\mathbf{x})$ and $\mathbb{E}[\|\nabla f(\mathbf{x}, \xi) - \nabla f(\mathbf{x})\|^2] \leq \sigma^2$.

3.2 Main Convergence Theorem

Let $\{\mathbf{x}_t\}_{t=0}^T$ be the sequence generated by the stochastic gradient descent algorithm with a fixed step size $\eta = \frac{1}{L}$. Then, the expected function value satisfies:

$$\mathbb{E}[f(\mathbf{x}_T)] - f(\mathbf{x}^*) \leq \left(1 - \frac{\mu}{L}\right)^T (f(\mathbf{x}_0) - f(\mathbf{x}^*)) + \frac{\eta \sigma^2}{2\mu}.$$

Proof. The proof involves establishing a recursive inequality using the strong convexity and smoothness properties, and then applying the law of total expectation along with variance bounds. Due to space constraints, this is a summary. $\square$

3.3 Theoretical Properties

3.3.1 Stability Analysis

The algorithm is stable under the choice of step size $\eta = \frac{1}{L}$, ensuring that the updates do not overshoot given the smoothness bound.

3.3.2 Sensitivity to Hyperparameters

The convergence rate is sensitive to the choice of η . If η is too large, the algorithm can diverge, while if it is too small, convergence is slow. The optimal step size is inversely proportional to the smoothness parameter L .

3.3.3 Comparison to Baseline Methods

Compared to gradient descent, stochastic gradient descent (SGD) has a slower convergence rate due to gradient variance but benefits from reduced computational cost per iteration. In the strongly convex and smooth setting, SGD achieves a linear rate of convergence, which is competitive with full-batch gradient descent in high-dimensional data regimes.

3.4 Discussion

The convergence theory presented provides a rigorous basis for the use of stochastic gradient descent in optimizing strongly convex and smooth functions. The linear convergence rate achieved under these conditions highlights the algorithm's efficiency, especially when working with large datasets where computational resources are a constraint. Future work could explore adaptive step size techniques to further enhance convergence properties.

4 Convergence Proof

4.1 Preliminaries (Definitions and Lemmas)

Before diving into the main proof, let's establish some foundational definitions and lemmas that will be crucial for our derivations.

4.1.1 Definitions

1. **Strong Convexity:** A function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is μ -strongly convex if for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{\mu}{2}\|\mathbf{y} - \mathbf{x}\|^2.$$

2. **Smoothness:** A function f is L -smooth if for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$:

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|^2.$$

3. **Gradient Noise:** The stochastic gradient $\nabla f(\mathbf{x}, \xi)$ satisfies:

$$\mathbb{E}[\nabla f(\mathbf{x}, \xi)] = \nabla f(\mathbf{x}), \quad \mathbb{E}[\|\nabla f(\mathbf{x}, \xi) - \nabla f(\mathbf{x})\|^2] \leq \sigma^2.$$

4.1.2 Lemmas

[Descent Lemma] For a L -smooth function f , the following holds for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$:

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|^2.$$

[Strong Convexity Inequality] For a μ -strongly convex function f , the following holds for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$:

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{\mu}{2}\|\mathbf{y} - \mathbf{x}\|^2.$$

4.2 Main Proof (Step-by-Step Derivation)

We aim to prove that the expected function value satisfies:

$$\mathbb{E}[f(\mathbf{x}_T)] - f(\mathbf{x}^*) \leq \left(1 - \frac{\mu}{L}\right)^T (f(\mathbf{x}_0) - f(\mathbf{x}^*)) + \frac{\eta\sigma^2}{2\mu}.$$

4.2.1 Step 1: Recurrence Relation

Consider the update rule of SGD:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \nabla f(\mathbf{x}_t, \xi_t).$$

Using the smoothness of f , we have:

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) + \nabla f(\mathbf{x}_t)^T(\mathbf{x}_{t+1} - \mathbf{x}_t) + \frac{L}{2}\|\mathbf{x}_{t+1} - \mathbf{x}_t\|^2.$$

Substitute $\mathbf{x}_{t+1} - \mathbf{x}_t = -\eta \nabla f(\mathbf{x}_t, \xi_t)$:

$$f(\mathbf{x}_{t+1}) \leq f(\mathbf{x}_t) - \eta \nabla f(\mathbf{x}_t)^T \nabla f(\mathbf{x}_t, \xi_t) + \frac{L}{2}\eta^2 \|\nabla f(\mathbf{x}_t, \xi_t)\|^2.$$

4.2.2 Step 2: Taking Expectations

Taking expectation with respect to the randomness in ξ_t :

$$\mathbb{E}[f(\mathbf{x}_{t+1})] \leq \mathbb{E}[f(\mathbf{x}_t)] - \eta \mathbb{E}[\|\nabla f(\mathbf{x}_t)\|^2] + \frac{L}{2}\eta^2 \mathbb{E}[\|\nabla f(\mathbf{x}_t, \xi_t)\|^2].$$

Using the variance bound:

$$\begin{aligned} \mathbb{E}[\|\nabla f(\mathbf{x}_t, \xi_t)\|^2] &= \mathbb{E}[\|\nabla f(\mathbf{x}_t) + (\nabla f(\mathbf{x}_t, \xi_t) - \nabla f(\mathbf{x}_t))\|^2] \\ &= \|\nabla f(\mathbf{x}_t)\|^2 + \mathbb{E}[\|\nabla f(\mathbf{x}_t, \xi_t) - \nabla f(\mathbf{x}_t)\|^2] \leq \|\nabla f(\mathbf{x}_t)\|^2 + \sigma^2. \end{aligned}$$

Substituting back:

$$\mathbb{E}[f(\mathbf{x}_{t+1})] \leq \mathbb{E}[f(\mathbf{x}_t)] - \eta \|\nabla f(\mathbf{x}_t)\|^2 + \frac{L}{2}\eta^2 (\|\nabla f(\mathbf{x}_t)\|^2 + \sigma^2).$$

4.2.3 Step 3: Simplifying the Inequality

Rearranging terms:

$$\mathbb{E}[f(\mathbf{x}_{t+1})] \leq \mathbb{E}[f(\mathbf{x}_t)] - \left(\eta - \frac{L}{2}\eta^2 \right) \|\nabla f(\mathbf{x}_t)\|^2 + \frac{L}{2}\eta^2\sigma^2.$$

Choosing $\eta = \frac{1}{L}$, we have:

$$\mathbb{E}[f(\mathbf{x}_{t+1})] \leq \mathbb{E}[f(\mathbf{x}_t)] - \frac{1}{2L} \|\nabla f(\mathbf{x}_t)\|^2 + \frac{\sigma^2}{2L}.$$

4.2.4 Step 4: Using Strong Convexity

From the strong convexity, we have:

$$\|\nabla f(\mathbf{x}_t)\|^2 \geq 2\mu(f(\mathbf{x}_t) - f(\mathbf{x}^*)).$$

Substituting:

$$\mathbb{E}[f(\mathbf{x}_{t+1})] \leq \mathbb{E}[f(\mathbf{x}_t)] - \frac{\mu}{L} (\mathbb{E}[f(\mathbf{x}_t)] - f(\mathbf{x}^*)) + \frac{\sigma^2}{2L}.$$

4.2.5 Step 5: Establishing the Recurrence

Let $\Delta_t = \mathbb{E}[f(\mathbf{x}_t)] - f(\mathbf{x}^*)$. Then:

$$\Delta_{t+1} \leq \left(1 - \frac{\mu}{L}\right) \Delta_t + \frac{\sigma^2}{2L}.$$

4.3 Rate Derivation (With Constants)

Solving the recurrence relation:

$$\Delta_t \leq \left(1 - \frac{\mu}{L}\right)^t \Delta_0 + \frac{\sigma^2}{2L} \sum_{k=0}^{t-1} \left(1 - \frac{\mu}{L}\right)^k.$$

The geometric series sum is:

$$\sum_{k=0}^{t-1} \left(1 - \frac{\mu}{L}\right)^k = \frac{1 - \left(1 - \frac{\mu}{L}\right)^t}{\frac{\mu}{L}}.$$

Thus:

$$\Delta_t \leq \left(1 - \frac{\mu}{L}\right)^t \Delta_0 + \frac{\sigma^2}{2\mu}.$$

4.4 Special Cases

4.4.1 Case 1: No Gradient Noise ($\sigma^2 = 0$)

In this case, the convergence rate simplifies to:

$$\Delta_t \leq \left(1 - \frac{\mu}{L}\right)^t \Delta_0.$$

This indicates a linear convergence rate without any variance.

4.4.2 Case 2: Large Step Size

If $\eta > \frac{1}{L}$, the algorithm may diverge due to the violation of the smoothness condition. Hence, the choice $\eta = \frac{1}{L}$ is critical for stability.